Biology Assessment

Class 10 • Human Biology & Plant Biology

Instructions: Answer all questions. The answer sheet used for testing may contain answers out of order.

Q1 **2 marks**

Define photosynthesis and write its word equation.

Q2 **2 marks**

Which organelle is primarily involved in photosynthesis? Name the pigment mainly responsible for absorbing light.

Q3 **3 marks**

Explain the role of chloroplasts in photosynthesis. Mention the two major stages of the process.

Q4 **4 marks**

Describe the flow of blood through the human heart starting from the right atrium and ending at the aorta. Include the names of the major valves crossed.

Q5 **3 marks**

Draw a labelled diagram of an alveolus showing the capillary, air space, and direction of gas exchange.

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Q6 **4 marks**

Describe the pathway of food through the human digestive system and explain where most nutrient absorption occurs. Mention the roles of the stomach, small intestine, liver and pancreas.

Q7 **5 marks**

Draw and label a nephron. Include Bowman's capsule, glomerulus, proximal tubule, loop of Henle, distal tubule and collecting duct.

Q8 **5 marks**

Explain two structural differences between palisade mesophyll and spongy mesophyll and relate each difference to its function.

Q9 **4 marks**

Describe transpiration in plants and name two environmental factors that can increase its rate.

Q10 **3 marks**

Explain how the structure of xylem vessels facilitates water transport. Mention at least two structural features.

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Q11(a) **2 marks**

Two potted plants are kept under different light conditions. Plant A has bright light and broad green leaves; Plant B has dim light and pale, elongated leaves. Explain one likely effect on photosynthesis.

Q11(b) **3 marks**

Suggest one practical measure that could help Plant B recover and explain why it would help.

Q12 **2 marks**

A resting person has a tidal volume of 0.5 L per breath and breathes 12 times per minute. Calculate the total ventilation per minute.

Q13 **2 marks**

If physiological dead space is 0.15 L per breath, calculate the alveolar ventilation per minute using the values from Question 12. Show your working.